



# B' B<sup>3</sup>/<sub>4</sub> B± B° / Boba

Multi-tracker meta-search for qBittorrent â€” self-hosted, containerised, private-tracker-aware.

[Quick start](#) Â· [Tokens & keys](#) Â· [Documentation](#) Â· [Architecture](#) Â· [Testing](#) Â· [Contributing](#)

|              |                                                                       |                      |               |            |    |               |               |
|--------------|-----------------------------------------------------------------------|----------------------|---------------|------------|----|---------------|---------------|
| python tests | 5358 collected                                                        | frontend tests       | 371 collected | plugins    | 48 | merge service | FastAPI :7187 |
| ci           | manual (./ci.sh, no auto-trigger)                                     | pre-build invariants | 30            | challenges | 31 |               |               |
| scanners     | snyk   sonar   bandit   ruff   semgrep   trivy   gitleaks   pip-audit |                      |               |            |    | license       | Apache 2.0    |

## Features

- **Merge Search Service** â€” FastAPI service (:7187) that fans out across 40+ trackers, deduplicates results, streams via SSE.
- **Real-time results** â€” result\_found events arrive as each tracker completes, no blocking.
- **Private-tracker bridge** â€” Authenticated downloads via webui-bridge.py for RuTracker, Kinozal, NNM-Club, IPTorrents.
- **Freeleech-only IPTorrents** â€” Automation never costs ratio (see [constitution VIII](#)).
- **Opt-in quality stack** â€” SonarQube + Snyk + Semgrep + Trivy + Gitleaks + bandit + pip-audit behind docker-compose.quality.yml.
- **Opt-in observability** â€” Prometheus + Grafana dashboards behind the same profile system.
- **Hook system** â€” Register scripts via POST /api/v1/hooks for search / download events.
- **Angular 21 SPA dashboard** â€” dark-themed, signals-based, per-tracker status chips with CAPTCHA re-login, virtual-scroll-ready sort.
- **PWA-ready** â€” favicon + launcher icons + web manifest ship from frontend/public/.

- **Container-first** – rootless Podman or Docker, single `./start.sh` boot.
  - **Resource-pressure early-warning** – [challenges/scripts/resource\\_pressure\\_signature\\_challenge.sh](#) checks 5 independently-falsifiable host-pressure signatures (runaway RSS, thread-count vs `ulimit -u`, container EAGAIN cascades, swap pressure, cgroup memory pressure) so a forced-logout precursor is caught *before* `user@1000.service` gets SIGKILLED, not after. Wired into `scripts/pre_build_verification.sh` invariant 25 and a standing `boba-resource-pressure-check.timer` systemd-user unit. See [incident writeup](#).
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## Quick start

```
git clone https://github.com/milos85vasic/qBitTorrent.git
cd qBitTorrent
```

# Optional: configure tracker credentials + tokens

```
cp .env.example .env
```

```
$EDITOR .env # see "Tokens & API keys" below
```

```
./setup.sh # one-time
```

```
./start.sh # Angular build + containers + plugin sync
```

Then visit:

| URL                                                                                 | Purpose                                  | Auth          |
|-------------------------------------------------------------------------------------|------------------------------------------|---------------|
| <a href="http://localhost:7187/">http://localhost:7187/</a>                         | Merge-search dashboard (SPA)             | –             |
| <a href="http://localhost:7187/docs">http://localhost:7187/docs</a>                 | FastAPI Swagger UI                       | –             |
| <a href="http://localhost:7187/openapi.json">http://localhost:7187/openapi.json</a> | OpenAPI schema                           | –             |
| <a href="http://localhost:7186/">http://localhost:7186/</a>                         | qBittorrent WebUI proxy                  | admin / admin |
| <a href="http://localhost:7185/">http://localhost:7185/</a>                         | qBittorrent internal (container)         | –             |
| <a href="http://localhost:7188/">http://localhost:7188/</a>                         | WebUI bridge (private-tracker downloads) | –             |

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## Tokens & API keys

– **Configure any credentials you need in `.env` before running `./start.sh`.** The single source of truth is [docs/TOKENS\\_AND\\_KEYS.md](#) – which variable is mandatory, which is optional, and **where to register** for each one.

## Tracker credentials “ env vars AND cookies files (auto-loaded)

Boba supports **two mechanisms** for tracker auth, and both are respected together:

1. **Env vars in .env** “ <TRACKER>\_USERNAME / <TRACKER>\_PASSWORD / <TRACKER>\_COOKIES.
2. **Cookies files in ~/Downloads/** “ canonical naming cookies\_<tracker>.txt (lowercase), Netscape TSV format exported from a logged-in browser session.

The loader scripts/load-tracker-cookies.sh auto-invokes before every boba-svc up, boba-svc restart, install.sh Stage 6, and start.sh boot “ so refreshing a session is a one-step *browser re-export + restart*. Source directory defaults to \$HOME/Downloads, overridable via TRACKER\_COOKIE\_DIR or --dir <path>.

See the [Tracker Credentials Manual](#) for the per-tracker matrix, browser-export step-by-step, precedence rules, and the security posture (Â§11.4.10.A leak audit, cookie-value-never-logged). Common questions: [FAQ](#).

### At a glance

| Category              | Mandatory   | Optional                                    | Documentation       |
|-----------------------|-------------|---------------------------------------------|---------------------|
| qBittorrent WebUI     | “           | WEBUI_USERNAME, WEBUI_PASSWORD              | <a href="#">Â§1</a> |
| Private trackers      | per-tracker | “                                           | <a href="#">Â§2</a> |
| Public tracker uplift | “           | JACKETT_API_KEY, “                          | <a href="#">Â§3</a> |
| Metadata enrichment   | “           | TMDB_API_KEY, TVDB_API_KEY, “               | <a href="#">Â§4</a> |
| Security scanning     | “           | SNYK_TOKEN, SONAR_TOKEN, “                  | <a href="#">Â§5</a> |
| Observability         | “           | GRAFANA_USER, GRAFANA_PASSWORD              | <a href="#">Â§6</a> |
| Orchestrator tuning   | “           | ALLOWED_ORIGINS, MAX_CONCURRENT_TRACKERS, “ | <a href="#">Â§7</a> |

### Where to register (fast links)

| Provider  | Signup                                                                                          | Notes               |
|-----------|-------------------------------------------------------------------------------------------------|---------------------|
| RuTracker | <a href="https://rutracker.org/forum/register.php">https://rutracker.org/forum/register.php</a> | Username + password |
| Kinozal   | <a href="https://kinozal.tv/signup.php">https://kinozal.tv/signup.php</a>                       | May require invite  |
| NNM-Club  |                                                                                                 | Cookie-based auth   |

| Provider    | Signup                                                                                                      | Notes                           |
|-------------|-------------------------------------------------------------------------------------------------------------|---------------------------------|
|             | <a href="https://nnmclub.to/forum/ucp.php?mode=register">https://nnmclub.to/forum/ucp.php?mode=register</a> |                                 |
| IPTorrents  | <a href="https://iptorrents.com/">https://iptorrents.com/</a>                                               | Invite-only; freeleech enforced |
| Jackett     | <a href="https://github.com/Jackett/Jackett">https://github.com/Jackett/Jackett</a>                         | Self-hosted indexer             |
| TMDb        | <a href="https://www.themoviedb.org/signup">https://www.themoviedb.org/signup</a>                           | v3 auth API key                 |
| TVDb        | <a href="https://thetvdb.com/api-information">https://thetvdb.com/api-information</a>                       | Subscription API                |
| MusicBrainz | <a href="https://musicbrainz.org/doc/MusicBrainz_API">https://musicbrainz.org/doc/MusicBrainz_API</a>       | Free “ needs UA only            |
| Snyk        | <a href="https://app.snyk.io/">https://app.snyk.io/</a>                                                     | Account + Auth Token            |
| SonarCloud  | <a href="https://sonarcloud.io/account/security/">https://sonarcloud.io/account/security/</a>               | Or run the compose SonarQube    |
| Gitleaks    | <a href="https://gitleaks.io/">https://gitleaks.io/</a>                                                     | Optional commercial license     |

## Documentation

Everything the platform offers, indexed:

### Tracked-Items + Status Documents

The mandated entry point (§11.4.57 / §11.4.212) for every tracker and status document in this repository “ Last modified and Revision are sourced verbatim from each document’s own revision header (§11.4.44); a “ means the document currently carries no revision header or has no HTML/PDF export yet (never fabricated).

| Document                    | Last modified        | Revision | Markdown                 | HTML                 |
|-----------------------------|----------------------|----------|--------------------------|----------------------|
| Issues                      | 2026-08-18T19:17:52Z | 10       | <a href="#">Markdown</a> | <a href="#">HTML</a> |
| Issues_Summary              | “                    | “        | <a href="#">Markdown</a> | <a href="#">HTML</a> |
| Fixed                       | 2026-08-18T20:55:14Z | 20       | <a href="#">Markdown</a> | <a href="#">HTML</a> |
| Fixed_Summary               | “                    | “        | <a href="#">Markdown</a> | <a href="#">HTML</a> |
| CONTINUATION                | 2026-08-18T21:10:20Z | 24       | <a href="#">Markdown</a> | <a href="#">HTML</a> |
| PORTING-FROM-LAVA           | 2026-07-01T16:18:43Z | 1        | <a href="#">Markdown</a> | “ (n yet expor       |
| REMAINING_WORK_PLAN         | 2026-08-07T19:10:54Z | 2        | <a href="#">Markdown</a> | <a href="#">HTML</a> |
| GOVERNANCE_AUDIT_2026-08-07 | 2026-08-07T19:10:54Z | 1        | <a href="#">Markdown</a> | “ (n yet expor       |

| Document                                | Last modified        | Revision | Markdown                 | HTML                 |
|-----------------------------------------|----------------------|----------|--------------------------|----------------------|
| browser_extension/Status                | 2026-06-13T13:10:00Z | 15       | <a href="#">Markdown</a> | <a href="#">HTML</a> |
| browser_extension/<br>Status_Summary    | 2026-06-10T20:05:00Z | 3        | <a href="#">Markdown</a> | <a href="#">HTML</a> |
| browser_extension/<br>RELEASE_READINESS | 2026-06-13T13:10:00Z | 4        | <a href="#">Markdown</a> | <a href="#">HTML</a> |
| codegraph/Status                        | 2026-08-18T13:33:41Z | 2        | <a href="#">Markdown</a> | <a href="#">HTML</a> |
| COMPLETION_STATUS                       | 2026-06-06T00:00:00Z | 1        | <a href="#">Markdown</a> | <a href="#">HTML</a> |
| RELEASE_READINESS_20260616              | 2026-06-16T10:30:00Z | 1        | <a href="#">Markdown</a> | <a href="#">HTML</a> |
| features/Status                         | 2026-08-18T13:33:41Z | 8        | <a href="#">Markdown</a> | <a href="#">HTML</a> |
| features/Status_Summary                 | 2026-06-16T23:30:00Z | 7        | <a href="#">Markdown</a> | <a href="#">HTML</a> |
| QA_DISCOVERY_LEDGER                     | 2026-08-18T22:10:00Z | 9        | <a href="#">Markdown</a> | <a href="#">HTML</a> |

## Getting started & operation

- [docs/USER\\_MANUAL.md](#) â€” end-user walkthrough
- [docs/TOKENS\\_AND\\_KEYS.md](#) â€” **â€” credentials / tokens / env vars (mandatory vs optional + registration links)**
- [docs/DESIGN\\_SYSTEM.md](#) â€” runtime-switchable palette catalogue (8 palettes, dark + light, Darcula default)
- [docs/PLUGINS.md](#) â€” the 48 plugin engines
- [docs/PLUGIN\\_TROUBLESHOOTING.md](#) â€” what to check when a plugin breaks

## Architecture & subsystems

- [docs/architecture/](#) â€” 5 Mermaid diagrams (topology, search lifecycle, plugin execution, bridge, shutdown)
- [docs/DATA\\_MODEL.md](#) â€” Pydantic schemas & lifecycle (no relational DB)
- [docs/CONCURRENCY.md](#) â€” asyncio semaphore + TTL caches + retry/backoff
- [docs/OBSERVABILITY.md](#) â€” Prometheus + Grafana + OpenTelemetry
- [docs/PERFORMANCE.md](#) â€” benchmark + load + stress test layout
- [docs/api/openapi.json](#) â€” frozen OpenAPI spec (diffed in CI)

## Quality & security

- [docs/SECURITY.md](#) â€” threat model + credential storage
- [docs/SCANNING.md](#) â€” Snyk + Sonar + Semgrep + Trivy + Gitleaks + bandit + pip-audit
- [docs/QUALITY\\_STACK.md](#) â€” the opt-in docker-compose.quality.yml stack

## Incidents & QA evidence

Per Â§11.4.238, automated HelixQA coverage is the discovery layer â€” manual/operator/agent-found defects are confirmation-only, and every out-of-band finding gets a coverage-escape-audit entry here.

- [docs/QA\\_DISCOVERY\\_LEDGER.md](#) â€” the discovery-channel ledger + coverage-escape schema (also indexed above in Tracked-Items)
- [docs/incidents/](#) â€” host-safety + CONST-033 forensic investigations, including the latest [2026-08-18 2nd perceived forced-logout](#) triage (root cause: partial, Â§11.4.6 UNCONFIRMED: boundary honestly stated; preventive gate: the resource-pressure signature challenge above)
- [docs/qa/](#) â€” per-item captured machine evidence (Â§11.4.5 / Â§11.4.69 / Â§11.4.83), one directory per workable item / task run â€” e.g.Â [docs/qa/BOB-116/](#)

## Manual QA

Per Â§11.4.185, automated gates are necessary but not sufficient â€” every release deliverable requires a final human confirmation pass. Per-session manual-QA checklists live under [docs/manual-qa/](#), one file per session, in the Â§11.4.153 per-feature Status/video-confirmation format.

- [docs/manual-qa/2026-08-18-session-manual-qa-checklist.md](#) â€” 12-feature checklist covering the boba-stack reachability, cross-tracker search/dedup, download flow, BOB-112 /healthz DDoS mitigation, constitution v68 landing, resource-pressure timer, CONST-033 challenge, and pre-build gate for the 457cca4â†’63f3f88 session range

## Testing

- [docs/TESTING.md](#) â€” catalogue of every test type (30 rows)
- [docs/COVERAGE\\_BASELINE.md](#) â€” per-module coverage gates
- [tests/README.md](#) â€” layout of the tests/ tree

## Development

- [CONTRIBUTING.md](#) â€” contribution workflow
- [CLAUDE.md](#) â€” Claude Code agent protocol (TDD + rebuild-reboot)
- [AGENTS.md](#) â€” runtime development guidance
- [.specify/memory/constitution.md](#) â€” **binding architectural contract (v1.1.0)**
- [docs/scripts/](#) â€” per-script user guides (Â§11.4.18), one <name>.md per scripts/\*.sh / challenges/scripts/\*.sh â€” e.g.Â [commit-push-all.md](#), [capture-workable-items-db-delta.md](#), [regenerate-continuation-exports.md](#)

## Courses (self-paced, Asciiinema)

- [courses/01-operator/](#) â€” Your first search
- [courses/02-plugin-author/](#) â€” Authoring a nova3 plugin

- [courses/03-contributor/](#) "TDD + rebuild-reboot deep dive"
- [courses/04-security-ops/](#) "Threat model + scanner bundle"

## Release / CI

- [CHANGELOG.md](#)
- [releases/README.md](#) â€” release artefacts layout
- [docs/OUT\\_OF\\_SANDBOX.md](#) â€” items requiring external credentials (HelixQA / OpenCode / submodule orgs)

# Architecture

```

    http://:7186 â€” qbittorrent-proxy â€”
    â€” (python:3.12-alpine) â€”
    http://:7186 â€” Download proxy â€”
    Download proxy (:7186) â€”
    â€”
    http://:7187 â€” Merge search service (:7187)
    Angular SPA + â€” â€” /api/v1/search + SSE â€”
    FastAPI â€” â€” /api/v1/bridge/health â€”
    â€” â€” /api/v1/auth/... â€”
    â€” â€” â€”
    â€”
    http://:7188 â€” webui-bridge.py (host process) â€” private-tracker b
    ot-in (docker-compose.quality.yml):
    http://:9000 SonarQube profile: quality
    http://:9090 Prometheus profile: observability
    http://:3000 Grafana profile: observability
    (run-once) Snyk / Semgrep /
    Trivy / Gitleaks profile: run-once

```

Full Mermaid renders in [docs/architecture/](#).

## Testing

```
# Non-live-HTTP suites " run anywhere, ~10s
python3 -m pytest tests/unit/ tests/e2e/ tests/contract/ --no-cov -q

# Benchmarks " dedupe perf + live-HTTP fan-out benchmarks, ~4min
python3 -m pytest tests/benchmark/ --no-cov -q

# Security " hits the live merge service, needs stack up
python3 -m pytest tests/security/ --no-cov -q

# Integration " same
python3 -m pytest tests/integration/ --no-cov -q -m "not requires credenti
```

```
# Frontend
npx --prefix frontend ng test --watch=false

# Full scanner sweep (non-interactive; skips scanners with missing tokens)
./scripts/scan.sh --all

# Pre-build invariant sweep " 25 checks (workable-items integrity, docs-
# sync, resource-pressure signature, mutation-residue, "11.4.238 discover
# ledger freshness, ...) run before every build
./scripts/pre_build_verification.sh

# HelixQA Challenges " 31 scripts under challenges/scripts/, e.g.:
./challenges/scripts/resource_pressure_signature_challenge.sh
./challenges/scripts/ddos_resilience_challenge.sh --healthz
```

See [docs/TESTING.md](#) for the 30-row test-type catalogue.

---

## Releases

Non-interactive builder at [scripts/build-releases.sh](#) produces artefacts under [releases/<version>/](#):

```
./scripts/build-releases.sh                # all targets, all channels
./scripts/build-releases.sh frontend        # one target
./scripts/build-releases.sh --channel release
```

Each artefact ships with SHA256SUMS + BUILD\_INFO.json.

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## Contributing

See [CONTRIBUTING.md](#) and the TDD protocol in [CLAUDE.md](#). PRs must keep the following green:

- Python unit + e2e + contract (pytest " 4448 tests collected, see docs/TESTING.md)
  - Frontend Vitest (ng test " 371 tests collected, see docs/TESTING.md)
  - Ruff + bandit + shellcheck (via scripts/scan.sh)
- 

## License

Apache 2.0 " see [LICENSE](#).